Steel Genesis

Open Rails 1.6.1

Modlist & Guide

Version 1.6.1 — January 2026

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Installation Guide

Installing Open Rails 1.6.1

Install Open Rails the usual Windows way:

1. Download the installer, Open.Rails.1.6.1.Setup.exe.
2. Run the installer and follow the normal Windows setup steps.
3. Launch Open Rails for the first time. A "Content" form pops up — this is Open Rails' built-in Content system, and it is the tool used for every wave of this guide.

The Built-In Content System

Open Rails does not need a separate mod organiser. Its built-in Content system installs, organises, and filters your add-ons, and it is the tool used for every wave of this guide.

The catalogue lets you filter content in two ways: by type — Routes, Train cars, or Train sets — and by terms of use — Free, Donation, or Commercial. Every route used in this guide is a Free route.

Installing from the catalogue is a self-installing download: you highlight the item, confirm where it should go, and Open Rails downloads and installs it for you. The detail pages for each wave describe the individual routes; this section describes the install process itself.

Installing a Route from the Catalogue

1. Open the Content form (it pops up on first launch).
2. Highlight the route you want — for example Demo Model 1 in Wave 0.
3. Set or confirm the "Install Path", the folder where content is installed.
4. Click "Install". Open Rails downloads and installs the route for you.
5. When the install completes, a dialog opens. Press OK, then press OK again to leave the Content form and reach the main menu.

The route is then ready to run. As a guide to the size involved, Wave 0's Demo Model 1 shows the catalogue's typical shape: roughly 272 MB to download and 330 MB installed, so allow around 600 MB of storage.

Installing Community Content Manually

Most community content ships as zip files rather than catalogue entries, so it is installed by hand:

1. Download the zip from the author's site — for example the Elvas Tower Download Library.
2. Extract the zip into the Content folder.
3. Start Open Rails and confirm the item appears in the Content form (and in the route list for routes).

That is all there is to it — Open Rails picks up community content placed in the Content folder.

Wave Installation Order

Install each wave's content when you start that wave, and follow the waves in order:

1. Wave 0 — The Rookie — installs Demo Model 1 only.
2. Wave 1 — The Road Freight — adds the BNSF Starter Route.
3. Wave 2 — The Yard — adds the Craven Timber Railway.
4. Wave 3 — The Limited — adds the Great Zig Zag Railway.

All four routes install through the Content form described above. Do not install later waves early — each wave's content is designed to build on the previous one, and each wave's sections assume its own route is the one in use.

Configuration (Quality of Life)

Wave 0 has no mods to install: every quality-of-life improvement there is configuration rather than content. Open Rails' built-in options cover it all:

- Arrange the F4 Track Monitor and F5 Train Driving Info windows where you want them; Open Rails remembers their positions between sessions.
- Choose fullscreen or windowed play with Alt+Enter.
- Open Rails' built-in settings govern the rest — graphics (renderer, fullscreen or windowed) and controls are all configured there.

Later waves build on this configured base, adding route content rather than further configuration.

Waves & New-Game Setups

Wave 0: The Rookie

Roleplaying Background

Welcome aboard. You are the newest hire at the fictional Aysgarth & North Riding Railway, a small line that has just been granted a single short diesel turn. The yard master has handed you the morning express: a British Rail 'blue'-era diesel working out of Edinburgh Waverley as far as Linlithgow — about twenty minutes of running on a short, forgiving main line. It is deliberately easy and exactly the right place to learn the craft of driving a train.

How to Play — Wave 0 (Beginner)

Installing Open Rails

Download and run the Open Rails 1.6.1 installer (Open.Rails.1.6.1.Setup.exe) and install it the usual Windows way. On first launch, a Content form pops up. Highlight the route Demo Model 1, set or confirm the Install Path, and click Install — the route needs roughly 600 MB of storage (the catalogue lists 272.42 MB to download and 329.99 MB installed). When the download and install finish, a dialog opens: press OK, then OK again to leave the Content form and reach the main menu.

Full step-by-step install and Content-system configuration is in the Installation Guide section of this document.

Driving Your First Train

From the main menu, click Start. A Windows Defender Firewall prompt may appear — allow access for Private networks. Open Rails starts fullscreen, and the simulation begins paused, with the Pause menu visible. Press Esc to begin driving.

You are now sitting in the driver's seat of a diesel locomotive, and the engine is already running. Press F5 to open the Train Driving Info window — the important lines are Speed, Direction, and Train Brake. Press F4 to open the Track Monitor, which shows track info, signals, and the speed limit; drag it where you like and Open Rails remembers window positions between sessions.

Drive with the mouse: move the pointer over the throttle handle and the train-brake handle on screen and click to move them — this is easier than keyboard control. Watch the speed limit on the Track Monitor and use the train brake to slow down and stop.

Camera & Cab Views

Press 1 for the driver's-seat view and 2 for the external (outside) view. In the external view, use the arrow keys to move around and hold the right mouse button while using the mouse wheel to look around. Press v to toggle the wipers. Use Alt+Enter to toggle between fullscreen and windowed, and Alt+F4 to stop the simulation and return to the main menu.

Config-as-QoL (Wave 0 Philosophy)

In Wave 0 there are no mods to install: every "quality of life" improvement comes from configuring Open Rails' own built-in options rather than installing content. Arrange the F4 Track Monitor and F5 Train Driving Info windows where you want them and let Open Rails remember the positions. Choose fullscreen or windowed play with Alt+Enter. Use the mouse on the in-cab handles instead of the keyboard. Open Rails' built-in settings cover the rest; later waves build on top of this configured base.

When You Are Ready for the Next Wave

Before moving on, check that you can:

- Start a session from the main menu and begin driving from the Pause menu.
- Move the throttle and the train brake with the mouse.
- Read Speed, Direction, and Train Brake on the F5 window.
- Keep the train within the speed limit shown on the F4 Track Monitor.
- Use the Track Monitor for track info, signals, and the speed limit.

When you can do all of that, you are ready for Wave 1.

Wave 0 — Modlist: Vanilla + Config-as-QoL

This wave uses only the built-in Content system and Open Rails' own settings. No external content is installed.

Routes

Demo Model 1

Version: Catalogue freeware — 2026

Dependencies: None — self-contained

Impact: Learning route for the Wave 0 tutorial. BR 'blue'-era diesel express from Edinburgh Waverley to Linlithgow (~20 minutes). Teaches cab controls, track monitor, speed limits and brake handling on a short, forgiving run.

Notes: Install via the Content form at first launch (see the Installation Guide). Set the Install Path before clicking Install. ~272 MB download / ~330 MB installed.

Wave 0 — Mechanics: How Trains Work

This wave introduces the handful of ideas every new driver needs before any content is added in later waves. None of them come from a mod — they are all part of how a train behaves in Open Rails itself.

Throttle and Train Brake

A locomotive is controlled by two big decisions: how much power you ask for and how much braking you apply. The throttle handle opens the engine's power, and the train-brake handle applies braking to the train as a whole. On screen you move both by clicking the mouse pointer on the handle — the same way you will drive the cab in every later wave.

Speed Limits

Every section of track has a speed limit. In Open Rails the limit is shown on the Track Monitor, and staying under it is the most important habit you can build now. On the Demo Model 1 run from Edinburgh Waverley to Linlithgow you have plenty of margin to practise picking up and dropping speed smoothly.

The Track Monitor

The F4 Track Monitor is your window on the track ahead: it shows track information, signals, and the speed limit. Signals tell you whether you may proceed, and reading them becomes second nature as quickly as the speed limit does.

Couplers and Consists

A train is a consist — a row of wagons, and on this route carriages, coupled together behind the locomotive. On Demo Model 1 the consist is already made up for you. Coupling and uncoupling wagons by hand is a real railroad skill, and it is saved for a later wave; for now you simply drive the train that is already behind you.

Next Steps

Master these basics — throttle, brake, speed limit, and the Track Monitor — and the short Demo Model 1 run gives you everything you need before Wave 1 brings real freight operations.

Wave 0 — Content: The Built-In Content System

Open Rails does not need a separate mod organizer. Its built-in Content system installs, organises, and filters your add-ons, and it is the tool used for every wave of this guide.

Content Types

The content catalogue organises add-ons into several types. The ones you meet in this guide are:

- Routes — the tracks and scenery you drive on. Demo Model 1, this wave's route, is a Routes item.
- Train cars — the individual locomotives and wagons you couple into a train.
- Train sets — ready-made groups of cars, often a locomotive plus a matched rake of wagons, ready to couple up and go.

Filters: Free, Donation, Commercial

The catalogue lets you filter content by its terms of use. Free items cost nothing. Donation items are free to use but ask you to support the author. Commercial items are paid. Every route used in this guide is Free.

What the Catalogue Holds

The catalogue currently lists self-contained routes only — each route bundles everything it needs (trains, activities, and setup files) so it can be installed and run with nothing else. Demo Model 1 is exactly that: a self-contained route, created by Making Tracks Ltd, donated by the company and cut down from a full-size commercial route, with installation instructions included. Later waves build on the same system.

Wave 0 — Graphics: Built-In Quality

Wave 0 installs no graphics mods at all. Demo Model 1 ships with its own scenery and lighting, so the look of the game is the look of Open Rails out of the box.

Everything you can improve visually in this wave comes from Open Rails' own built-in options rather than add-ons. You can run the sim fullscreen or in a window — Alt+Enter toggles between them — and the renderer offers its own quality settings, which you can adjust in general terms from Open Rails' built-in options without needing a mod.

One built-in feature worth knowing about is superelevation, Open Rails' simulation of how track is banked on curves. It is part of the default simulation and needs no add-on.

As in every other area of Wave 0, the point is to get comfortable with the built-in experience before changing anything. When later waves add graphics content, you will be able to appreciate the difference precisely because you already know what Open Rails looks and feels like on its own.

Wave 1: The Road Freight

Roleplaying Background
Your shortline training run is behind you. The yard master at the fictional Aysgarth & North Riding Railway had nothing but praise, and a word has been passed to the regional Class I road — fictional, like the line that taught you. You are now promoted to road-freight service: a through-freight engineer on a mountain-grade main line, moving manifest freight between yards on a fast, busy railroad. It is a step up in every way that matters — longer trains, tighter timekeeping, and signals that decide whether you keep moving or come to a stop.

Strategy — Main-Line Freight (Intermediate)

Reading Signals
On a main line, signals govern your movement: a clear signal lets you proceed at the line's speed, while other aspects tell you to expect a stop or slow down ahead. The Track Monitor from Wave 0 shows the next signal along with the speed limit, so you can read what is coming before you reach it. Approach every signal expecting to stop or slow until you can see what it actually says — that habit keeps you out of trouble more reliably than any single aspect.

Longer Consists
The BNSF Starter Route's trainsets are noticeably longer than the Wave 0 run. A longer train takes more distance and time to pick up speed, and it takes far more distance to stop. Braking distance grows with the size of the train, so begin slowing for a stop or a speed restriction earlier than feels natural. Let the train tell you what it needs instead of fighting it.

Throttle & Train-Brake Discipline
You still drive with the mouse on the handles, exactly as you learned on Demo Model 1. What changes is the discipline around them. A long freight has a lot of mass, and slamming the throttle or the brake forward makes that mass complain — the train bunches up and stretches out, and control suffers. Move the handles in smooth, deliberate steps, give the train time to respond, and it will stay stretched and steady underneath you.

Basic Timetables & Activities
The route's built-in activities give you a proper job: a start point, a destination, and a time window to run within. Follow the activity instructions — they tell you where you are starting, where you are expected, and what is being asked of you. You are no longer just keeping under a speed limit; you are keeping a schedule, which is the real shape of a road-freight engineer's day.

Wave 1 — Modlist: Road Freight
Adds the first real route plus a complete diesel-freight trainset and activities, all self-contained.

Routes

BNSF Starter Route

Version: Catalogue freeware — 2026

Dependencies: None — self-contained (includes trainsets and activities)

Impact: BNSF Scenic Subdivision, Pacific Northwest (USA). Modern diesel freight on a main line: signals, longer consists, throttle and train-brake discipline, and basic timetable running through the included activities.

Notes: Install via the Content form. ~598 MB download / ~894 MB installed. Pick a BNSF freight activity to start.

Train Sets

BNSF Starter Trainsets

Version: Included with BNSF Starter Route

Dependencies: BNSF Starter Route

Impact: Ships the diesel locomotives and freight cars needed by the route's activities — no separate download required.

Notes: Bundled with the route; no extra install step.

Wave 1 — Mechanics: Signals and Train Handling

Wave 1 brings the mechanics that define main-line railroading. None of them come from a mod — they are part of how every real railroad works, and Open Rails simulates them all.

Signals as Authority

A signal is a direct order to the engineer. At its simplest, a clear aspect says proceed, and every other aspect limits how far or how fast you may go — most commonly by telling you to be prepared to stop at the next signal, or to stop right there. The Track Monitor tells you what the next signal shows, which lets you plan your braking rather than reacting at the last moment.

Brake Handling on Grades

Freight railroads are built over mountains, and the BNSF Starter Route's main line is no exception. On a descending grade the train wants to run away from you, so you control it with the brakes — and on a long descent, steady, sustained braking is what keeps the train under control. Dynamic braking, the way a diesel uses its own traction motors to hold the train back, becomes a working tool here in a way it never needed to be on Wave 0's flat run.

Coupler Slack and Smooth Throttle

A freight consist is not one solid object; it is a string of cars joined by couplers with slack between them. As the train stretches and bunches, that slack moves through the train. A sudden throttle or brake change throws the slack hard against the cars, which is uncomfortable and can even damage equipment. Smooth throttle and brake transitions keep the slack travelling gently through the consist, which is why the discipline from the strategy section matters so much on a longer train.

Consist Length and Stopping

The longer the consist, the more mass is moving — and the more energy the brakes must absorb. Where Wave 0's short express could stop on a casual braking application, a long freight needs an early, deliberate slowdown. Reading the terrain and the signals ahead, and starting to brake for stops and restrictions long before you reach them, is the single biggest adjustment from the Rookie wave.

Next Steps

Get comfortable reading signals and handling the longer consist, and the activities will carry you the rest of the way into genuine road-freight running.

Wave 1 — Content: The BNSF Starter Route

Wave 1's content is a single self-contained package. The BNSF Starter Route, created by TrainSimulations.net and free to download, is a starter route that gets you onto the main line: it comes with trainsets and activities based on the BNSF Scenic Subdivision in the Pacific Northwest Region of the United States.

A Full Scenic Main Line

Where Wave 0's Demo Model 1 was a short, flat lesson, the BNSF Starter Route is a real main line through real geography — the Scenic Subdivision of the Pacific Northwest, with the grades and scenery that come with mountain railroading. It is exactly the kind of route where signals matter, where long trains earn their keep, and where the timetable starts to feel like a job.

Bundled Diesel Power and Freight Cars

The route ships its own trainsets — the diesel locomotives and freight cars you need to run it — so there is no separate download and no extra install step. The trainsets are the modern diesel-freight equipment of a big North American railroad: multiple-unit diesel power up front and a long rake of freight cars behind. They are made up ready to run in the route's activities.

A Set of Activities

Alongside the route and trainsets come activities, ready-made jobs that give you a start point, a destination, and a time window. Pick a BNSF freight activity to begin. Because everything is bundled together, installing the route is all you need to do — the activities and equipment are right there, waiting.

Self-Contained by Design

The point of the BNSF Starter Route is that it asks for nothing else. No separate train-set download, no extra scenery, no dependencies. Install it through the Content form and every piece of Wave 1 is in place. That self-contained design is also the catalogue's general approach, and it is what keeps this wave — and every wave — simple to set up.

Wave 1 — Graphics: Built-In, As Before

Wave 1 adds no external graphics mods. The BNSF Starter Route ships its own scenery and cabviews, so the look of the Pacific Northwest is the route's own work — nothing extra to install and nothing to configure.

At this stage, Open Rails' built-in options remain the only graphics controls. Everything you learned to use in Wave 0 — the built-in quality settings, running fullscreen or in a window — still applies exactly as before. The route simply gives you more to look at: long freight consists, signal masts, and mountain grades, all rendered with the same out-of-the-box engine you already know.

That is deliberate. The point of the early waves is to know the baseline thoroughly. When later waves consider graphics content, you will recognise what the built-in experience looks like — because Wave 1 keeps it unchanged.

Roleplaying Background

The main-line runs are behind you, and the fictional railway's management has decided where you are most useful. You are promoted to yard foreman and posted to a remote timber branch, a little sawmill line where the whole of the day's working rests on two saddle-tank steam locomotives and a handful of timber wagons. Your job is no longer to take a train from one place to another, but to make it up and split it apart: putting empty wagons under the loading point at the sawmill, pulling the loaded ones out, and building and breaking the consists that the branch lives on. It is slower, more deliberate railroading — and it is where you learn to feel every wagon you are handling.

Strategy — Switching Operations (Advanced)

Coupler Work

Switching is mostly coupler work. You move up to a wagon at little more than walking pace so that the couplers meet and engage, then make sure they are properly locked together before you take any strain. Pulling away with a coupler that never fully engaged means dragging the wagon a few feet and then stopping dead when it lets go — at best an inconvenience, at worst a damaged coupling and a derailed car. The discipline is simple: couple slowly, check the engagement, and only then pull.

Runaround Moves

Most of the yard work depends on the runaround, the move that lets a locomotive get to the other end of its train. When there is no loop or runaround track — which is common on a branch like this — you pull past the train, uncouple, reverse back along the parallel track beside it, and couple onto the other end. It looks elaborate, but it is the basic grammar of making up and splitting trains, and you will do it over and over. Learning to read the track layout so you know when a runaround is possible, and to plan the sequence of moves before you start, is half of yard craft.

Industry Spotting

The sawmill only works if the wagons arrive in the right place. Spotting is the art of placing an empty wagon precisely at the loading point and pulling the loaded one away when it is full. Load and unload only happen when the wagon is properly in position, so getting the stop right matters: too short and the mill can't reach the wagon, too long and it hangs over into the runaround. Judge your speed so you roll to a stop with the wagon where the work is — that exactness is what separates a yard foreman from someone just moving trains.

Working at Reduced Speed

Everything in the yard happens slowly. A wagon being shoved into position has no tolerance for speed — a mistake at speed means a derailment, a crushed buffer, or a coupler torn off. Short, deliberate movements, patience between each one, and a steady hand on the controls are the whole of the job. The route rewards that patience: there is no timetable pressure in the yard, only the pressure you put on yourself to do the move cleanly the first time.

Wave 2 — Modlist: The Yard

Adds a small industrial route built around switching and industry work.

Routes

Craven Timber Railway

Version: Freeware — 2026 (site version)

Dependencies: None — self-contained

Impact: 5.8-mile NSW timber tramway from The Glen to a sawmill at Craven Station. Two saddle-tank steam locomotives (PWD32 and 529X), tight 120-metre curves, and six Wards River crossings. Small enough to learn switching, runaround moves and industry spotting.

Notes: Install via the Content form. Download page on the creator's site.

Wave 2 — Mechanics: Coupling, Uncoupling and Yard Work

Wave 2 brings the mechanics of the yard — the physical work of joining and separating cars that the Wave 1 road-freight runs never asked you to do. Like the signals and brakes of the previous wave, none of it comes from a mod; it is how real railroads work, and Open Rails simulates it.

Coupling and Uncoupling

Moving a car anywhere begins with coupling and uncoupling. To pick up a wagon you approach it slowly, let the couplers come together and engage, then back off gently and take the strain. To leave a wagon behind you uncouple it, then pull away and leave it standing. It is a deliberate, low-speed business: done carefully, you can make up and break up a whole train without anyone hurrying anything.

Reversing Moves

The yard is full of back-and-forth. A locomotive rarely just runs forward from A to B; it shoves wagons, pulls past, backs down alongside, and works from either end of its train. Being comfortable running in reverse — looking where the movement is going, judging your speed on the move back, and stopping in exactly the right place — is half of switching. The runaround move from the strategy section depends entirely on it.

Handbrakes

A car left standing in the yard must be held there. Handbrakes on the wagons hold them in place once they are uncoupled, and they matter on any bit of grade the branch throws at you. Setting and releasing them is part of every uncouple and every pickup: you want the wagon you are leaving to stay put, and the wagon you are taking to roll freely once you couple on.

The Feel of a Small Steam Locomotive

The two saddle-tank engines on the Craven line are small, and that changes how the train feels. There is no heavy mass of diesel and road-train inertia to mask things; you feel the train through the throttle and the brake directly. The engine responds to the handle almost immediately, and so does the consist behind it. That sensitivity makes you a better judge of speed — which is exactly what reduced-speed yard work demands.

Next Steps

Couple, shove, spot, hold. Get those four movements smooth, and the mechanics of the yard stop being something you think about and start being something you just do.

Wave 2 — Content: Craven Timber Railway

Wave 2's content is another single self-contained package. The Craven Timber Railway, created by Peter Newell and free to download, is a 5.8-mile / 9.6 km branch of the North Coast line in New South Wales, Australia. This typical timber 'tramway' carried timber out of The Glen to a sawmill adjacent to Craven Station, built to light railway standards with tight 120-metre curves, crossing the Wards River six times as it winds up the picturesque river valley.

A Complete Small Branch Route

The Craven line is a finished route in itself — a timber branch with the river valley around it, the station at Craven, and the sawmill at the heart of the operation. Where Wave 1 gave you a main line to run end to end, this route gives you a compact world built around one kind of work: getting timber wagons from the forest to the mill and back. Its size is a strength — short enough to learn, detailed enough to come back to.

A Working Sawmill Industry

The sawmill is not scenery; it is the reason the branch exists. Empty wagons are spotted at the loading point and loaded ones pulled away, which is exactly the industry work the strategy section practises. The route's whole rhythm — make up the empty wagons, run them to the mill, spot them, take the loaded ones back — is built around it.

Timber Tramway Character

Built to light railway standards, the Craven line has the character of a genuine timber tramway: tight curves, a winding river valley, and hard-won gradients. The six Wards River crossings and the 120-metre curves are part of the route's personality — they keep speed modest and make the careful, reduced-speed handling of the yard the natural way to run the whole branch.

Two Saddle-Tank Steam Locomotives

The route is worked by two saddle-tank steam locomotives, PWD32 and 529X. They are the branch's entire motive power, and they are exactly the kind of small, direct engines the mechanics section describes — sensitive through the throttle and brake, and perfectly suited to the stop-and-start rhythm of switching.

Self-Contained and Suited to Switching Practice

Like every route in this catalogue, the Craven Timber Railway asks for nothing else: no separate trainsets, no extra scenery, no dependencies. Install it through the Content form and the whole wave is in place. What it gives you in return is practice ground — a small, self-contained branch that exists to teach switching, runaround moves, and industry spotting, with enough depth to keep working it long after the moves become second nature.

Wave 2 — Graphics: The Route's Own Scenery

Wave 2 adds no external graphics mods. The Craven Timber Railway ships its own river-valley scenery, so the look of the branch is the route's own work — the Wards River, the timber-framed structures, and the bushland the line winds through. Nothing extra to install and nothing to configure. The built-in options remain the only graphics controls, exactly as in Waves 0 and 1. The point of the early waves still holds: know the baseline thoroughly before anything changes it. Wave 2 simply gives the engine a different world to render — a slower, more detailed one, where the scenery sits close to the track and rewards looking at it.

Roleplaying Background

After your yard apprenticeship the fictional railway trusts you with the mountain division's named steam service — the "Limited". It is a demanding schedule over the Blue Mountains, where punctuality and steam management decide the day: carry a full head of steam up the ruling grades, hold the timing through the zig-zag, and bring the train into Bowenfels on the minute. It is the difference between driving a train and running a named service.

Strategy — The Limited (Expert)

Steam Handling Basics

A steam locomotive runs on a head of steam — pressure in the boiler that the fire produces. The regulator, or throttle, admits that steam to the cylinders, and the cut-off controls how long it is admitted on each stroke. Together they decide how much effort the engine puts out and how much steam it consumes. On the flat you can run on a short cut-off and a partly-open regulator, saving steam; when you need power you open up. Let the pressure fall too far and the engine simply runs out of puff — running out of steam on a hill means stopping, and on a mountain grade that is the last thing you want.

Gradients & Momentum

The Blue Mountains do not climb gently, and a steam engine does not find power by magic. Keep your momentum up on the approach to a climb — a train that hits a grade with speed is a train that clears it; a train that stalls on it may not get moving again. Know where the ruling gradient sits on your section and plan the run so the hardest part is met with a full head of steam and a settled speed. Momentum is your friend going up and your enemy going down — the descending side of the route needs just as much care as the climb.

Reversing Zig-Zag Moves

The route's signature move is the zig-zag: the line climbs by running forward to a switchback, reversing, and running forward again on the next leg. Each reversal is a full stop and a change of direction on a grade, so the whole move lives or dies on accurate stopping. Stop at the right point and the run onto the next leg is straightforward; over-run or under-run and you are fighting the gradient with a train that is not where you meant it to be. Learn to read the boards and bring the train to rest exactly where the move reverses.

Timetable & Station Work

A named service runs to time. Stop at the right platforms, mind your scheduled time, and watch for other traffic on the single main line — the mountain division is one railway, and the Limited is not the only train on it. The tutorial activities build these skills progressively, so by the end you are running the whole route on the timetable, not just making the train move.

Wave 3 — Modlist: The Limited

Steam-era capstone: a mountain-engineering-marvel route with progressive steam tutorials.

Routes

Great Zig Zag Railway

Version: Freeware — 2026 (site version)

Dependencies: None — self-contained

Impact: Blue Mountains NSW heritage line from Mt Victoria toward Bowenfels, west of Lithgow. An international engineering marvel with reversing zig-zag moves and steep gradients, plus 7 progressive tutorial activities that teach steam locomotive handling step by step.

Notes: Install via the Content form. ~210 MB download. Play the tutorial activities in order.

Wave 3 — Mechanics: Steam, Grades and the Zig-Zag

Wave 3 brings the mechanics of steam-era mountain railroading — the physical business of raising steam, spending it wisely, and getting a heavy train up gradients that switchbacks were built for. As in the earlier waves, none of it comes from a mod; it is how real steam railways work, and Open Rails simulates it.

Building and Using a Head of Steam

Everything a steam locomotive does begins in the boiler. The fire heats the water, the water makes steam, and the steam under pressure is the engine's entire stock of energy. Raising a good head of steam takes time and attention; spending it wisely takes judgment. Open the regulator wide at every hill and the pressure falls; work on a short cut-off where you can and the pressure holds. The art of steam handling is balancing what the boiler makes against what the cylinders use — and knowing that an empty boiler is a stopped train.

The Regulator and Cut-Off

Two controls shape how the engine uses its steam. The regulator (throttle) admits steam from the boiler to the cylinders, and the cut-off decides how long each piston stroke gets steam before it is left to its own momentum. A long cut-off and a wide regulator give maximum effort and maximum consumption — the climbing combination. A short cut-off uses less steam per stroke, the economical running mode for level stretches. Finding the right pair for the gradient and speed ahead of you is the daily skill of every steam driver.

Banking and Helpers on Steep Grades

Not every mountain grade can be climbed with one engine alone. The prototype answer to that problem is the banking engine — a helper that couples on at the foot of the grade, shoves from the rear until the summit, and returns light. It is a familiar idea from mountain railway history, and it applies whenever a single engine lacks the tractive effort for a steep pitch. Keeping the train together and matching the helper's effort is a coordination skill all its own.

Why the Tutorials Are Ordered Progressively

The route's 7 tutorial activities are deliberately arranged in order of difficulty, and the order is part of the teaching. Each activity introduces the challenges of managing a steam-hauled train in steps — the basics first, then the harder moments of the mountain — so that by the later activities you have already practised the moves they ask of you. Working through them in order is the intended way to learn the route, exactly as the earlier waves built skill on skill.

Wave 3 — Content: Great Zig Zag Railway

Wave 3′s content is the wave's namesake: the Great Zig Zag Railway, created by Peter Newell and free to download. It is part of the Great Western line that connects Sydney with Western New South Wales, carried over the Blue Mountains west of Sydney between Mt Victoria and Bowenfels, west of Lithgow.

The Great Western Line Over the Blue Mountains

The route is a railway's answer to a wall of mountains. The Great Western line needed to cross the Blue Mountains, and the Zig Zag was the most cost-effective solution to do it in the late 19th century — an international engineering marvel by any measure of its day. The line's history is the whole reason the route exists: a hard-won crossing, built to a demanding standard, that every run over it re-enacts.

The Engineering Marvel: Zig-Zag Ladders

The route's name names its signature. The line climbs the mountainside not by a single brutal grade but by running forward on one ledge, reversing on a switchback, and running forward again on the next. The reversing zig-zag turns one steep mountain face into several gentler climbs, and it gives the route a character no other wave has — movement that is as much about careful stops and reverses as it is about running.

Seven Progressive Steam Tutorial Activities

The route comes with 7 tutorial activities that introduce you progressively to the challenges of managing a steam-hauled train. They are the wave's real content: each one teaches steam handling on the mountain a step at a time, so the wave works as a course rather than a single route. Play them in order, and Wave 3 becomes the steam mastery the whole progression has been building toward.

Self-Contained and Ready Through the Content System

Like every route in this catalogue, the Great Zig Zag Railway asks for nothing else — no separate trainsets, no extra scenery, no dependencies. Install it through the Content form and the wave is in place. The download is around 210 MB, and what it delivers is the capstone: a mountain route, its engineering marvel, and a full course in running steam trains.

Wave 3 — Graphics: The Route's Own Scenery

Wave 3 adds no external graphics mods. The Great Zig Zag Railway brings its own mountain-valley scenery and steam-era character — the Blue Mountains climbing west of Sydney, the ledges the line follows up the valley side, and the heritage look of an engineering marvel from the late 19th century. Nothing extra to install and nothing to configure.

The built-in options remain the only graphics controls, exactly as in Waves 0, 1 and 2. By now the baseline is thoroughly known, and the point still holds: Wave 3 simply gives the engine a grander world to render — a mountain crossing where the scenery is the whole point of the route, and where the switchbacks give you views back along the line as you climb.

Glossary

Cab
The driver's seat — the compartment from which the engineer controls the locomotive, shown by pressing 1 in the cab view.

Consist
A set of coupled vehicles — locomotives, wagons and carriages — forming a train.

Content form
The built-in catalogue and installer that pops up when Open Rails first launches. Every route in this guide is installed through it.

Coupler
The mechanism that joins vehicles together. Coupling brings the couplers together and engages them; uncoupling releases them.

Handbrake
The parking brake on an individual car, used to hold a wagon in place once it has been uncoupled and left standing.

HUD
Head-up display — the on-screen overlay of driving information. In this guide the F5 Train Driving Info window is the HUD.

Industry spotting
Placing a car precisely at an industry — such as the Craven Timber Railway's sawmill — so it can be loaded or unloaded.

Install Path
The folder, set or confirmed in the Content form, where content is installed.

Limited
A named premium passenger service. Wave 3's "The Limited" is a named steam service running over the Blue Mountains.

Runaround
Moving the locomotive around its train — pulling past it, uncoupling, and coupling onto the other end — so it can push from behind.

Saddle tank
A small steam locomotive whose water tank straddles the boiler. The Craven line's PWD32 and 529X are saddle tanks.

Signal
An aspect that governs movement, telling the engineer to proceed, to be prepared to stop at the next signal, or to stop right there.

Superelevation
The banking of the track on curves, which lets trains hold higher speeds through them. Open Rails simulates it as part of the default simulation.

Throttle / regulator
The control admitting power to the wheels on a diesel, or steam to the cylinders on a steam locomotive. The regulator is the steam-engine name for the same control.

Timetable
The scheduled running plan a train is meant to follow. The Limited runs to a timetable; the yard does not.

Track monitor
The F4 window showing track information, signals and the speed limit.

Train brake
The brake controlling the whole train, applied from the cab, as opposed to a single car's handbrake.

Yard / switching
Low-speed car movement operations — coupling, uncoupling, shunting, runaround moves and industry spotting.

Zig-zag
A track alignment that reverses direction to climb, running forward on one ledge, reversing on a switchback, and running forward again on the next. The Great Zig Zag Railway's signature move.